

Anup Shrestha

Full-Stack Software Engineer | AI Engineer

anupstha0168@gmail.com • +977 9846900168 • [GitHub](#) • [LinkedIn](#)

SUMMARY

Full-Stack Software Engineer with experience building production-grade web applications and enterprise software solutions for banking, telecommunications, and pharmaceutical organizations. Skilled in React.js, Next.js, Node.js, Express.js, NestJS, LoopBack, PostgreSQL, Redis, RabbitMQ, and microservice architecture. Experienced in developing enterprise AI applications using LLM Integration, LangChain, LlamaIndex, RAG, AI Agents, and Speech-to-Text (STT)/Text-to-Speech (TTS). Passionate about building scalable, reliable software that solves real business problems.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python

Frontend: React.js, Next.js, Redux, Tailwind CSS

Backend: Node.js, Express.js, NestJS, LoopBack, REST APIs, Microservices

Databases: PostgreSQL, MongoDB, MySQL, Redis, Firebase

AI & Automation: LLM Integration, LangChain, LlamaIndex, AI Agents, RAG, Speech-to-Text (STT), Text-to-Speech (TTS), n8n, ActivePieces, Windmill, Python OCR

Vector Databases: Weaviate, Chroma

Tools & DevOps: Git, GitHub, Docker, Kafka, RabbitMQ, Prisma, TypeORM, Nginx, Portainer, Vercel, Postman

WORK EXPERIENCE

Software Engineer · Amnil Technology

(Sept 2024 – Present)

- Delivered production-grade full-stack and AI solutions for enterprise clients including Smart Axiata, NMB Bank, Machhapuchchhre Bank, SBL Bank, and Quest Pharmaceuticals.
- Developed scalable full-stack applications using React.js, Next.js, Node.js, Express.js, NestJS, PostgreSQL, Redis, Kafka, RabbitMQ, and microservice architecture. Optimized large data queries and implemented asynchronous job queue systems to improve application performance and scalability.
- Built enterprise AI applications through LLM Integration using LangChain, LlamaIndex, RAG, AI Agents, vector databases, and Speech-to-Text (STT)/Text-to-Speech (TTS).
- Automated enterprise workflows using Python OCR and Windmill, solving complex business problems while contributing throughout the software development lifecycle from requirement analysis to deployment and production support.

Full-Stack Developer Intern · Ramailo Tech

(Aug 2023 – Dec 2023) [Link](#)

- Developed and delivered a production-grade restaurant management platform for Japanese clients using React.js, Node.js, Express.js, and MySQL.
- Collaborated with senior developers in an Agile environment, contributing to feature development, code reviews, testing, deployment, and code quality improvements.

KEY PROJECTS

Smart Axiata AI Chatbot

Tech Stack: React.js | Express.js | LangChain | RAG | AI Agents | PostgreSQL | Weaviate | STT/TTS

- Developed and deployed a production AI chatbot for Smart Axiata, supporting over 50,000 daily users by delivering intelligent customer support through text and voice interactions.
- Built with LangChain, RAG, AI Agents, vector databases, and STT/TTS to deliver accurate, context aware responses from enterprise knowledge sources.

Loan Automation Platform

Tech Stack: Next.js | LoopBack | PostgreSQL | TypeORM | Python OCR | Windmill | LLM Integration

- Developed a loan automation platform for NMB Bank and SBL Bank that automates the processing and analysis of Audited, Provisional, and Projected financial reports.
- Automated OCR-based data extraction, report generation, and secure document delivery using Python OCR and Windmill workflows.

Machhapuchchhre Bank (MBL) Internal AI Assistant

Tech Stack: React.js | Express.js | LangChain | LlamaIndex | RAG | Chroma | PostgreSQL

- Developed an internal AI assistant that enables employees to instantly access attendance records, HR policies, staff details, loan policies, bank information, and internal documents through natural language queries.
- Implemented LangChain, LlamaIndex, and RAG pipelines to provide fast, accurate, and context aware enterprise knowledge retrieval.

Quest Pharmaceuticals AI Chatbot

Tech Stack: Next.js | Express.js | LangChain | LlamaIndex | PostgreSQL | Weaviate | Kafka

- Developed an AI-powered chatbot to help customers and employees quickly search pharmaceutical products and retrieve product information.
- Built using LangChain, LlamaIndex, RAG and Weaviate to provide accurate, context aware responses from the enterprise knowledge base.

NMB Internal AI Chatbot

Tech Stack: React.js | Express.js | LangChain | RAG | AI Agents | Vector Database | PostgreSQL | RabbitMQ

- Developed an internal AI assistant that enables employees to search NRB directives, HR policies, operational procedures, and internal documentation through natural language conversations.
- Designed AI Agent workflows and RAG pipelines to deliver accurate, context aware enterprise knowledge retrieval.

EDUCATION

Bachelor of Information Technology (BIT)

(2020 – 2024)

KIST College, Purbanchal University — Kathmandu, Nepal

ACHIEVEMENTS & CERTIFICATIONS

- 2nd Place — Intra College IT Project Competition — Awarded for best project among competing college teams
- 2nd Place — Kathmandu University IT Meet — Competed and placed among top university-level IT projects [Link](#)

